

2019 ACL

A Multiscale Visualization of Attention in the Transformer Model

Jesse Vig

Palo Alto Research Center

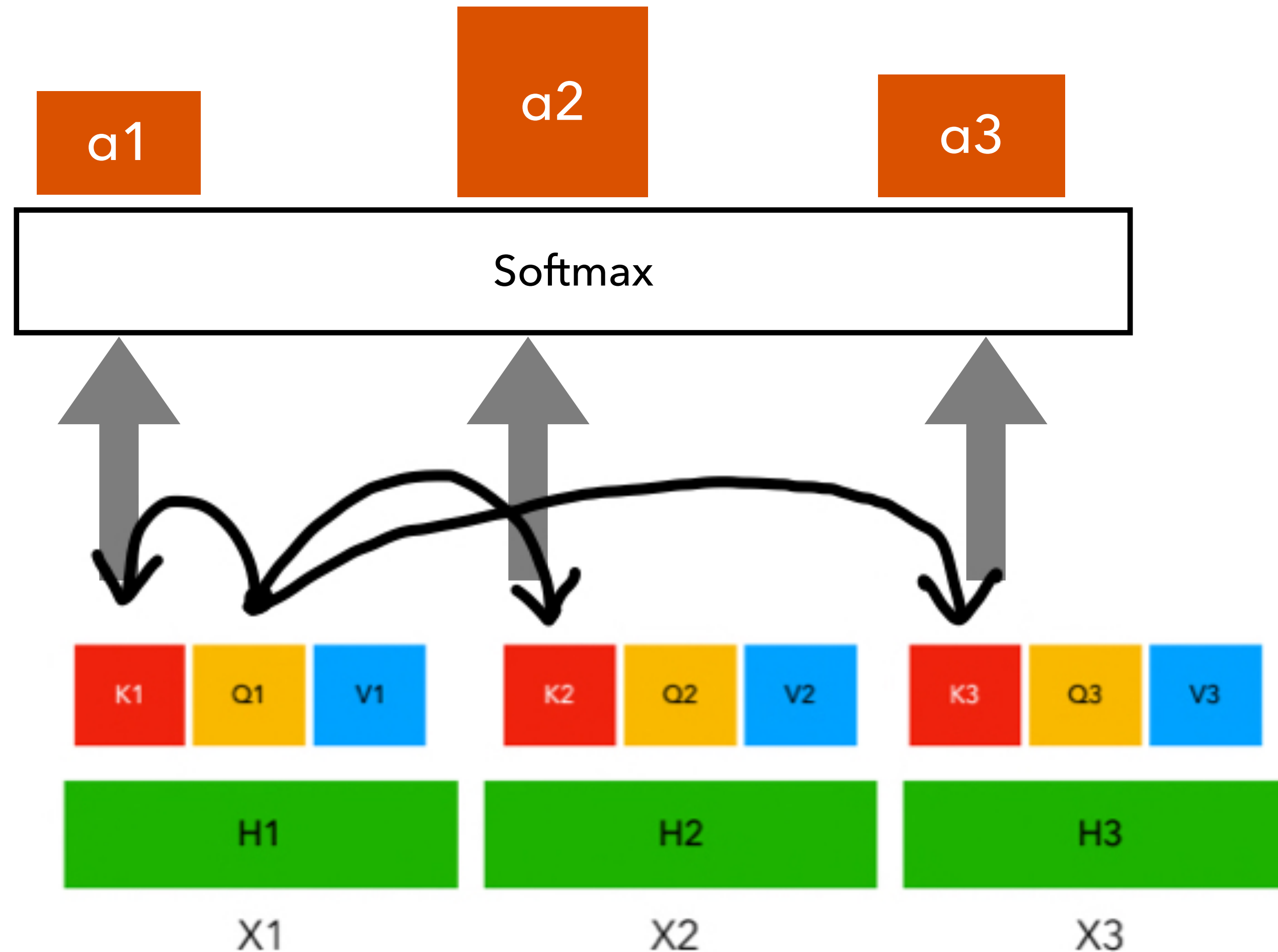
딥러닝 논문 읽기 모임
자연어처리팀 : 백지윤, 진명훈
발표자 : 백지윤

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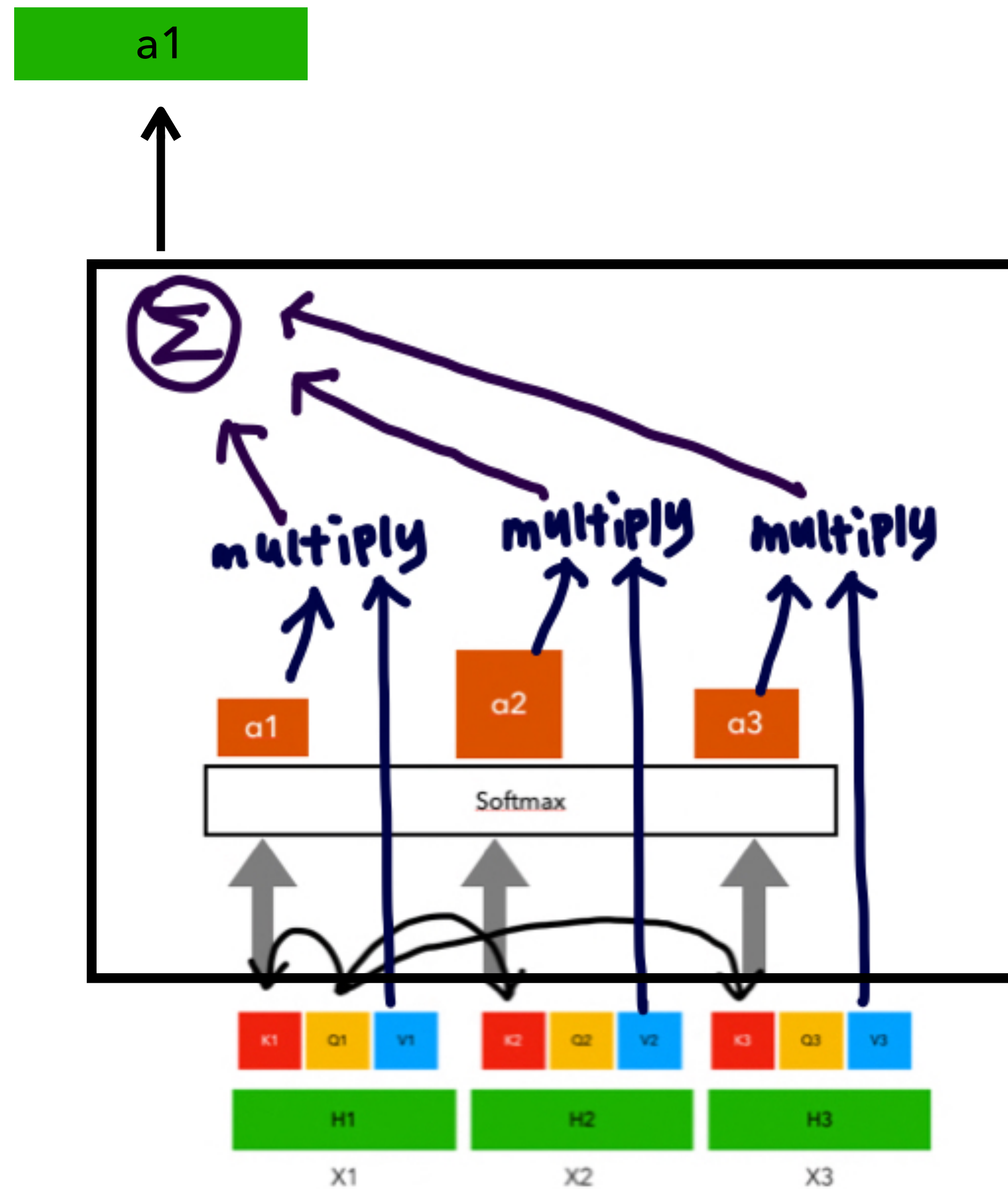
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1. Introduction - Transformer

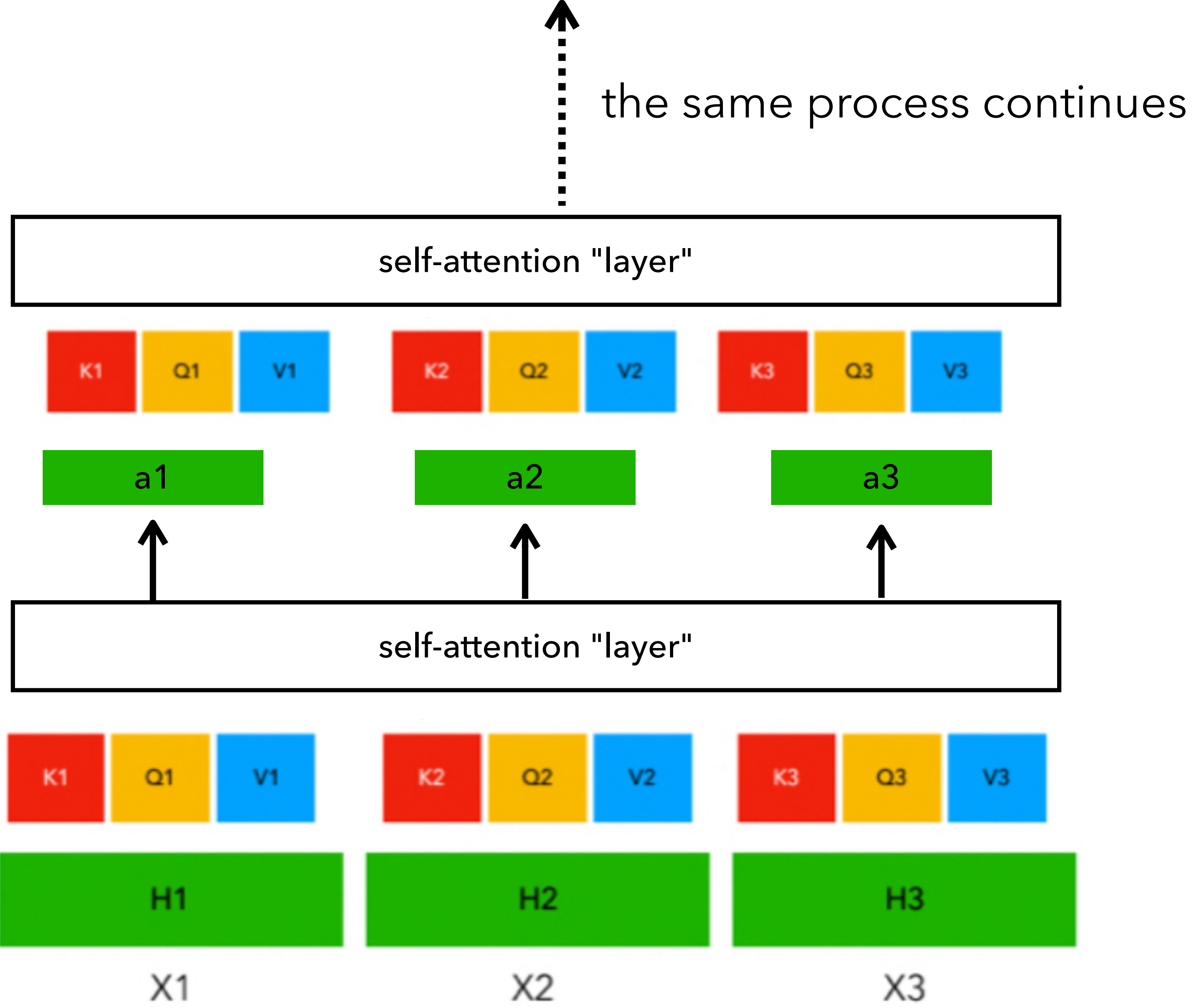
Transformer's key principle - Self-Attention



1. Introduction - Transformer

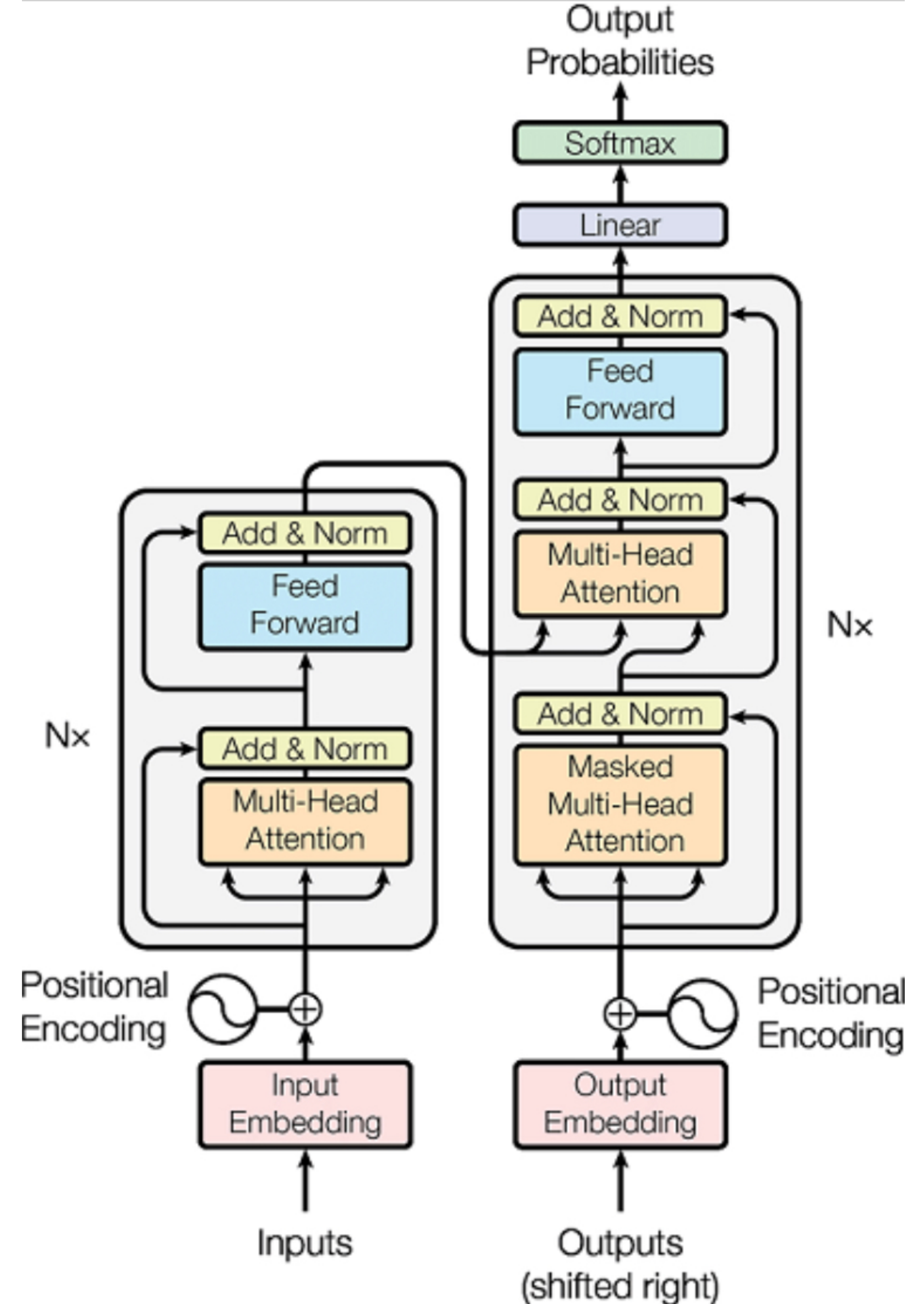


self-attention
"layer"



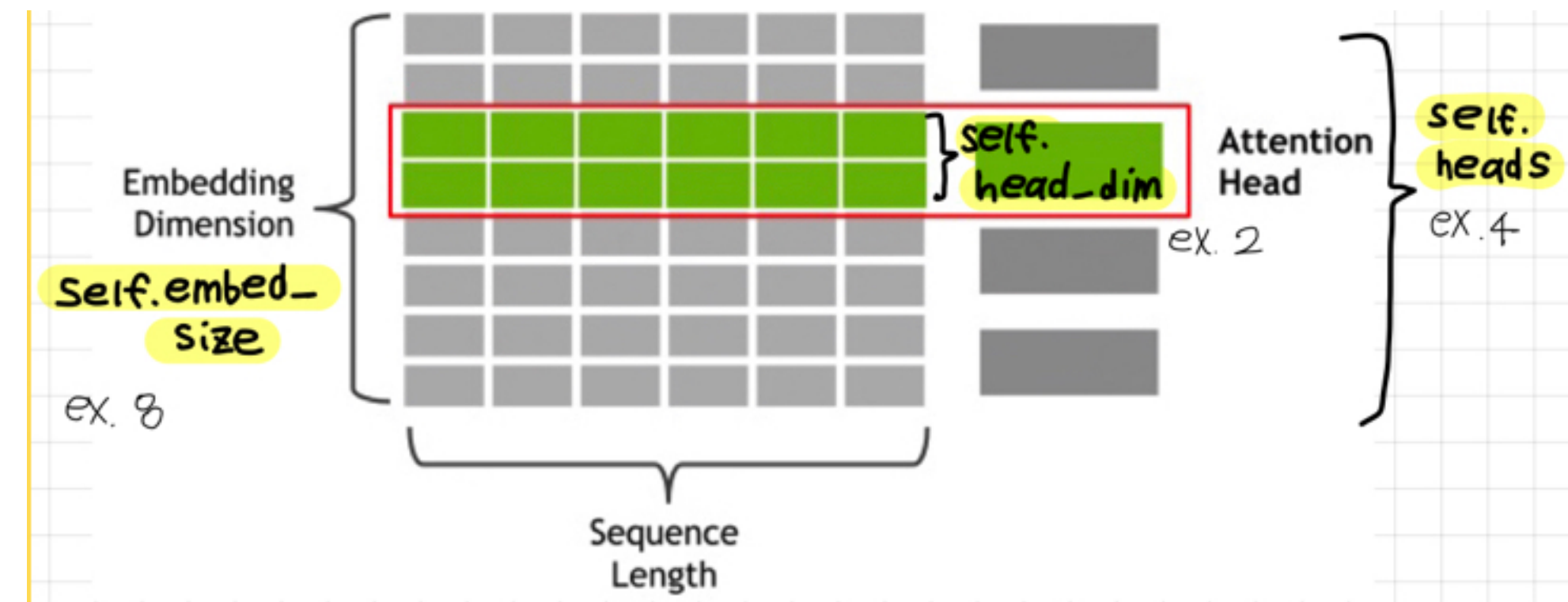
Transformer

- Actual Transformer assigns many heads per each word rather than just one head as mentioned before
- A decoder's key and value vectors come from an encoder
- There are some other detailed stuff to talk about (positional encoding etc.) I will go over it later on for myself.....🙄



Transformer

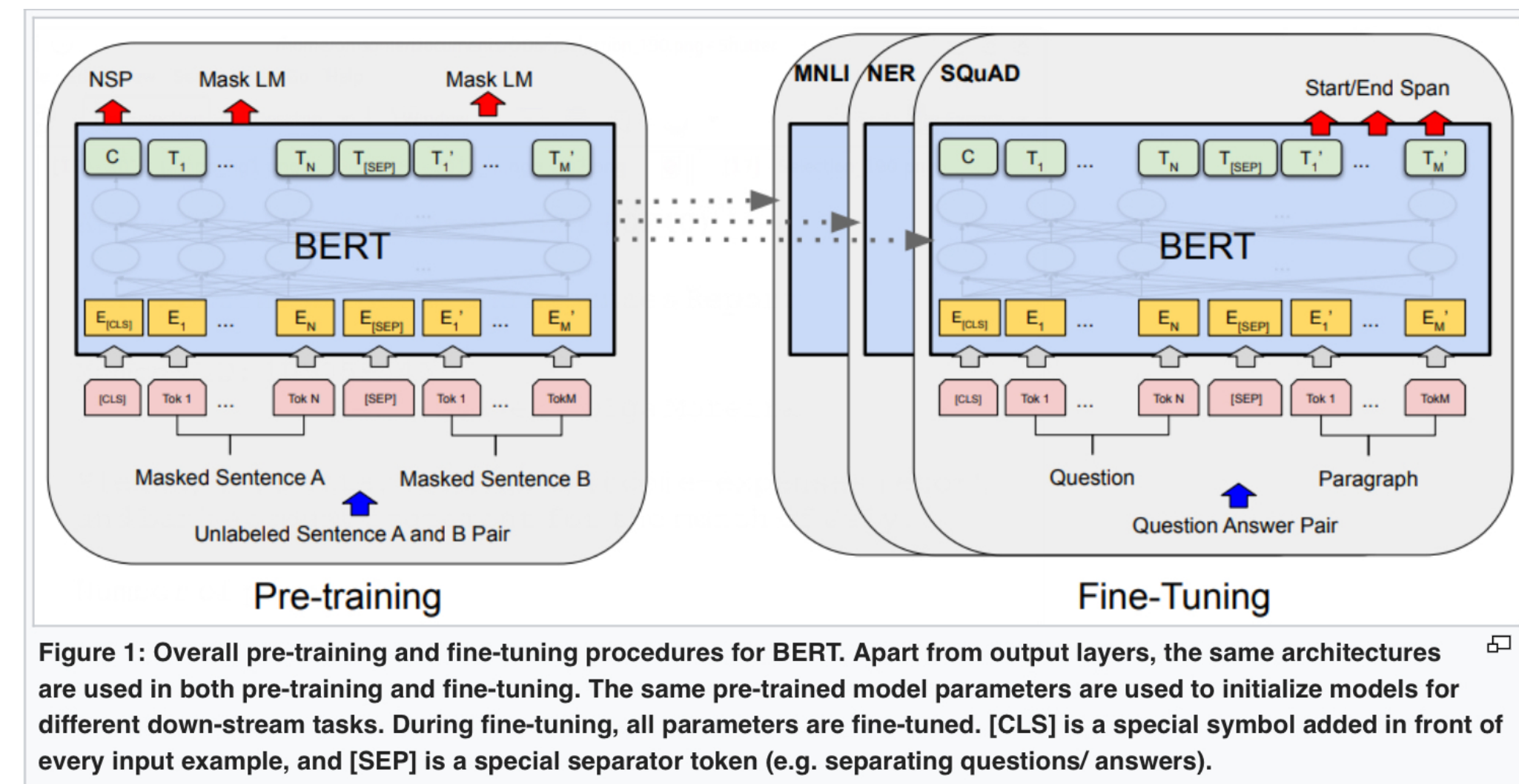
- for example, if the embedding dimension of each word is 8, a sequence length is 3, the number of heads is 4, then a final shape will be (N,3,4,2)
- A decoder's key and value vectors come from an encoder



```
enc_src=self.encoder(src,src_mask)
out=self.decoder(trg,enc_src,src_mask,trg_mask)
#codes inside Decoder>
def forward(x,enc_out,enc_out...)
```

Bert

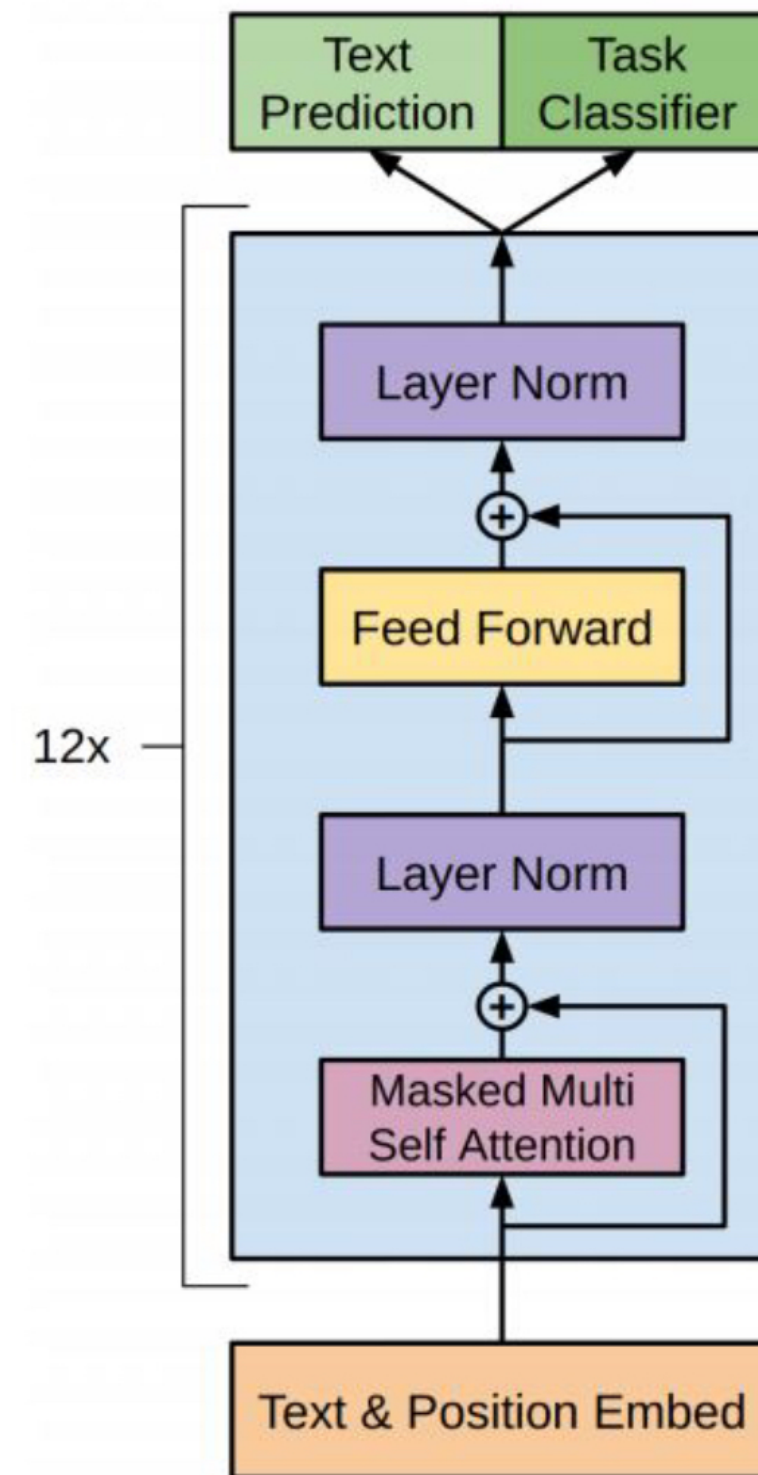
- Bert ; Transformer Encoder + Fully connected layer
- In order to be a smart language model, Bert completes two difficult tasks at a same time (MLM & NSP)
- After acquiring a linguistic ability, Bert is ready for fine-tuning.



[CLS] I want to be a [Mask] [Sep]
Tomorrow will be rainy.

GPT

- GPT ; Transformer Decoder + Fully connected layer
- In order to be a fluent language model, GPT gets through one important task



나는 학교에 간다

GPT - 2

<START> 나는 학교에

```
print(generate_sent("이때", gpt_model, greedy=True))
>> "이 때문에 일부 전문가들은 ... "
```

2. Why Visualization Tool & challenges

- An advantage of using attention is that it can help interpret a model by showing how the model assigns weight to different input element **through visualization**
- One challenge for visualizing attention in the Transformer is that it uses a multi-layer, multi-head attention mechanism. Ex) 24 layers and 16 heads -> $24 * 16 = 384$ unique attention structures already ! 🌀

Attention-head View

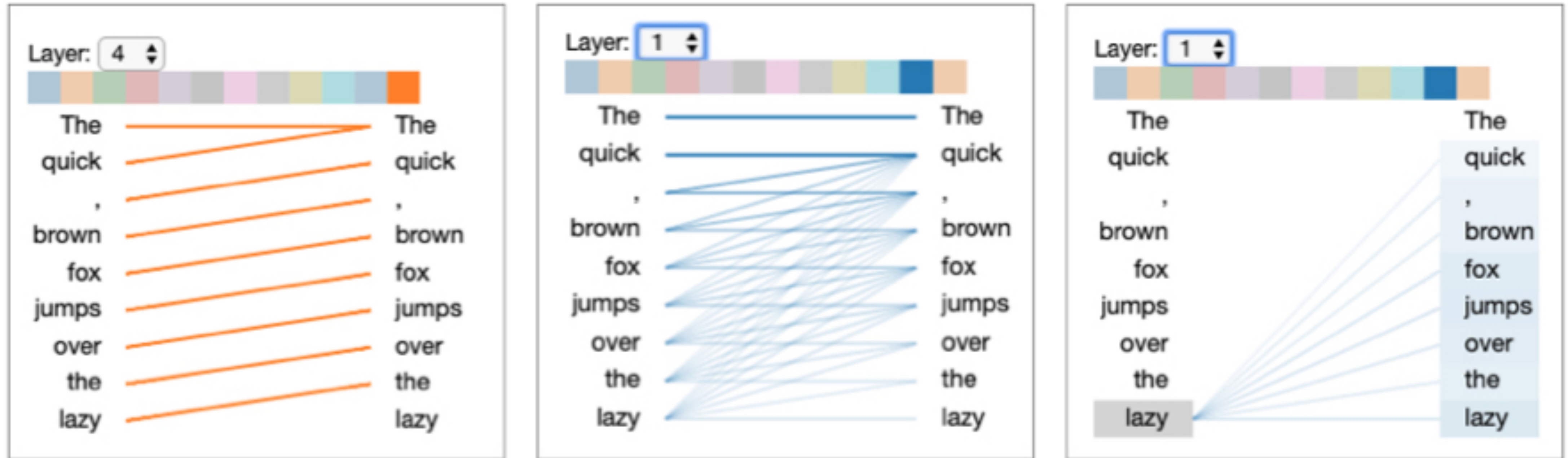


Figure 1: Attention-head view for GPT-2, for the input text *The quick, brown fox jumps over the lazy*. The left and center figures represent different layers / attention heads. The right figure depicts the same layer/head as the center figure, but with the token *lazy* selected.

Attention-head View

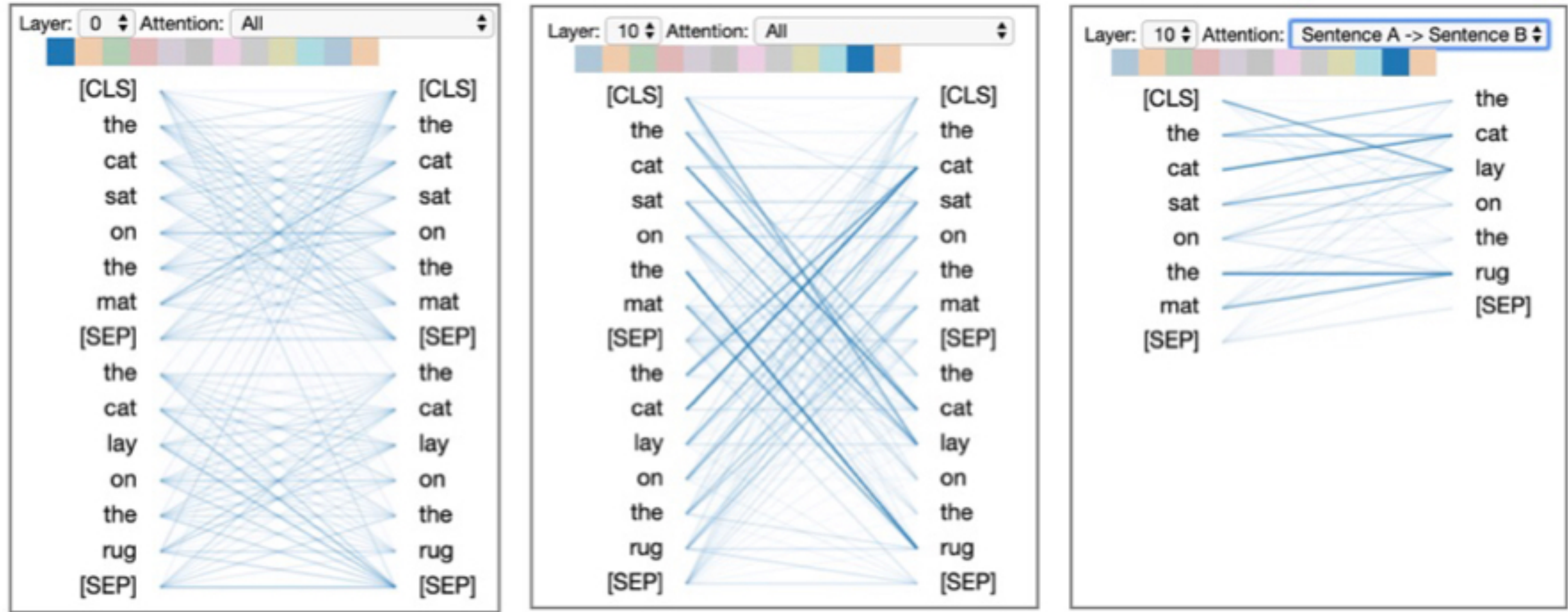


Figure 2: Attention-head view for BERT, for inputs *the cat sat on the mat* (Sentence A) and *the cat lay on the rug* (Sentence B). The left and center figures represent different layers / attention heads. The right figure depicts the same layer/head as the center figure, but with *Sentence A* $\rightarrow$ *Sentence B* filter selected.

Attention-head View

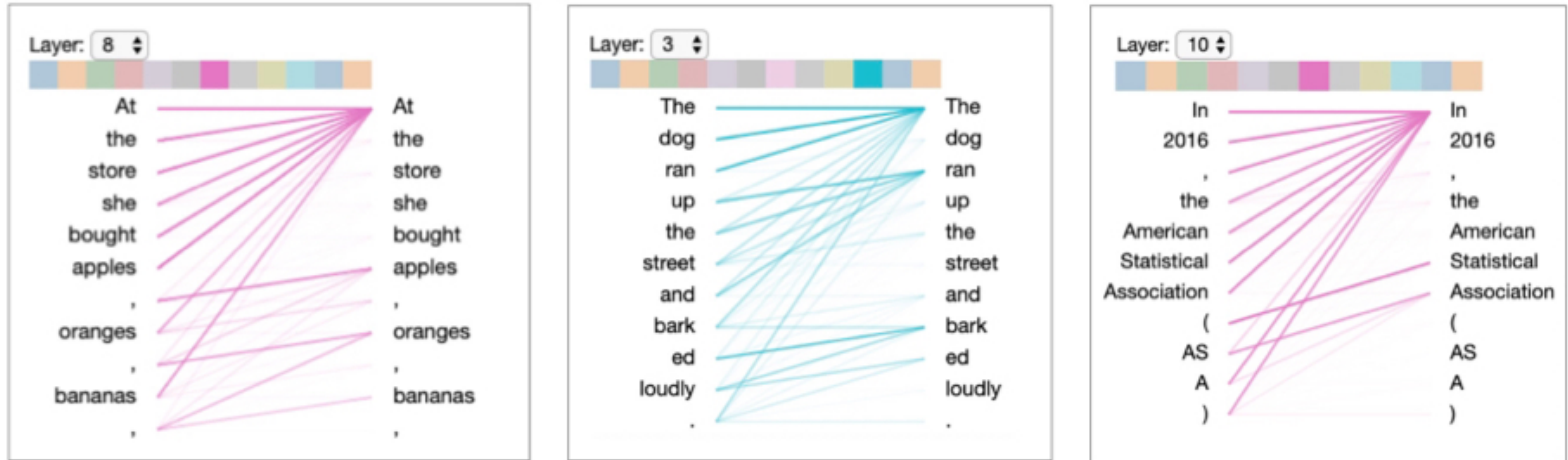


Figure 3: Examples of attention heads in GPT-2 that capture specific lexical patterns: list items (left); verbs (center); and acronyms (right). Similar patterns were observed in these attention heads for other inputs. Attention directed toward first token is likely null attention ([Vig and Belinkov, 2019](#)).

Use Case : Detecting Model Bias

The doctor asked the nurse a question.
He **asked her if she ever had a heart attack.**

The doctor asked the nurse a question.
She said **"I'm not sure what you're talking about."**

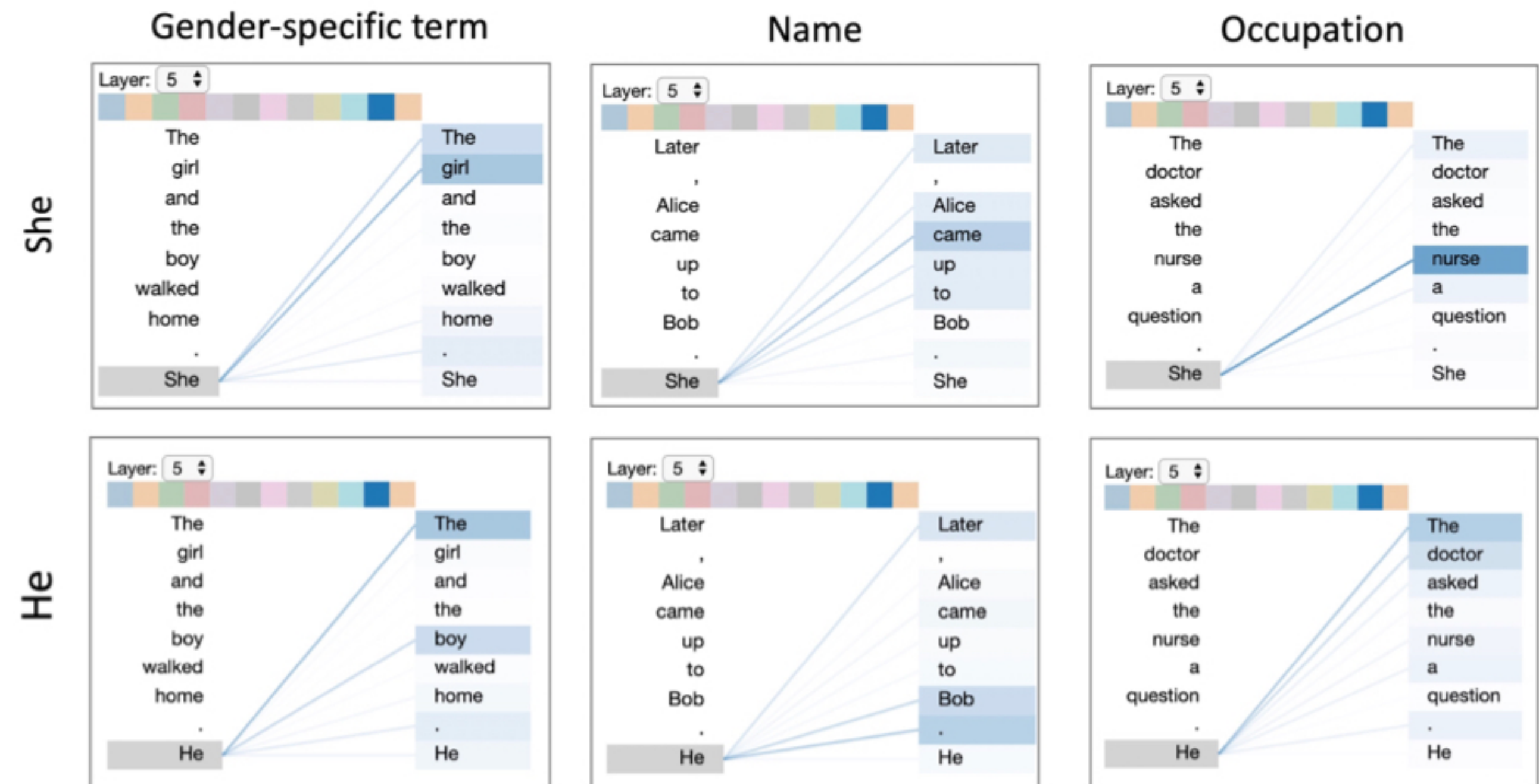


Figure 4: Attention pattern in GPT-2 related to coreference resolution suggests the model may encode gender bias.

Model View

Model View can be especially useful for paraphrase detection task.

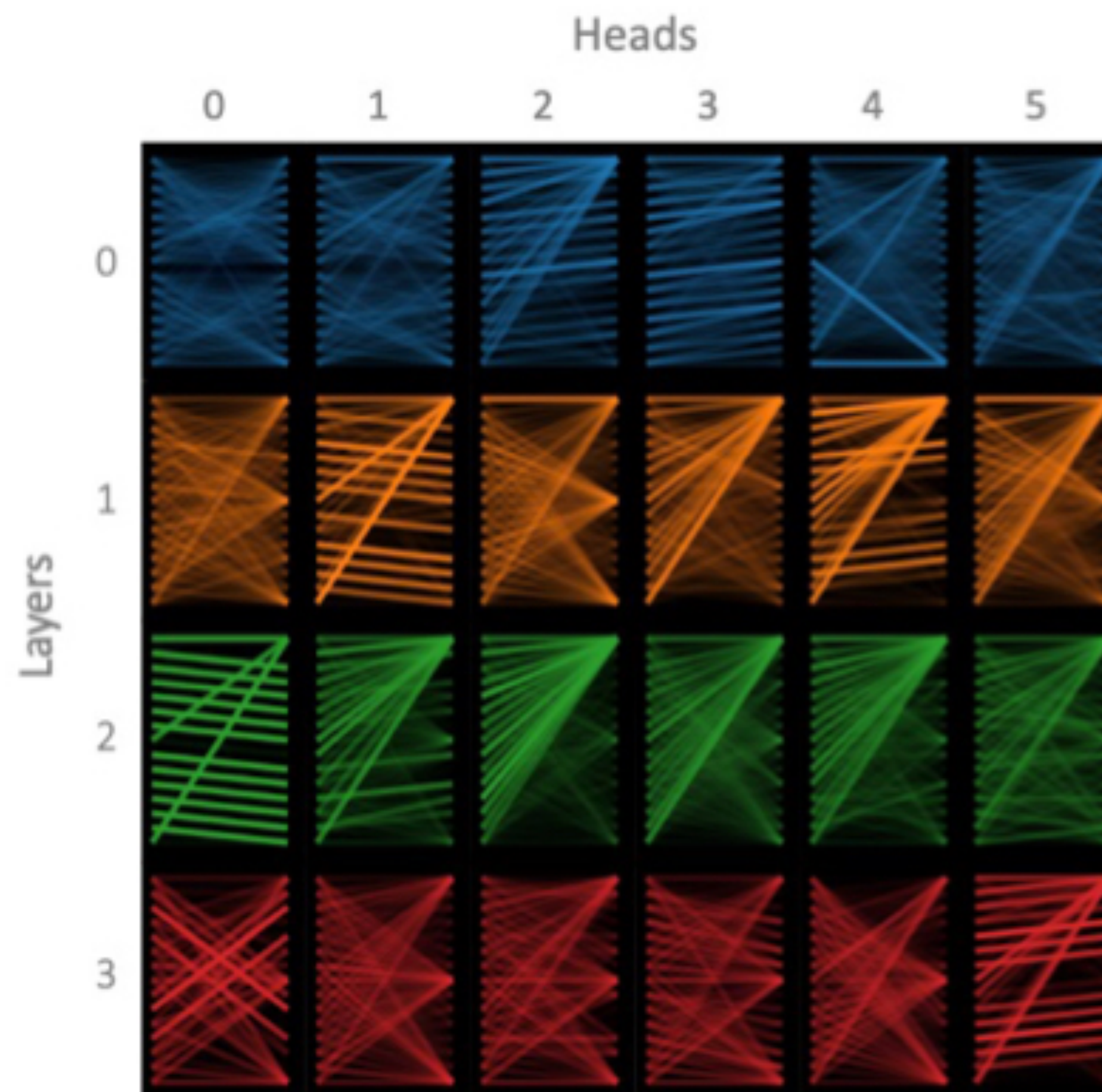


Figure 5: Model view of BERT, for same inputs as in Figure 2. Excludes layers 4–11 and heads 6–11.

Neuron View

Positive : Blue , Negative : Orange
Color saturation : magnitude of value

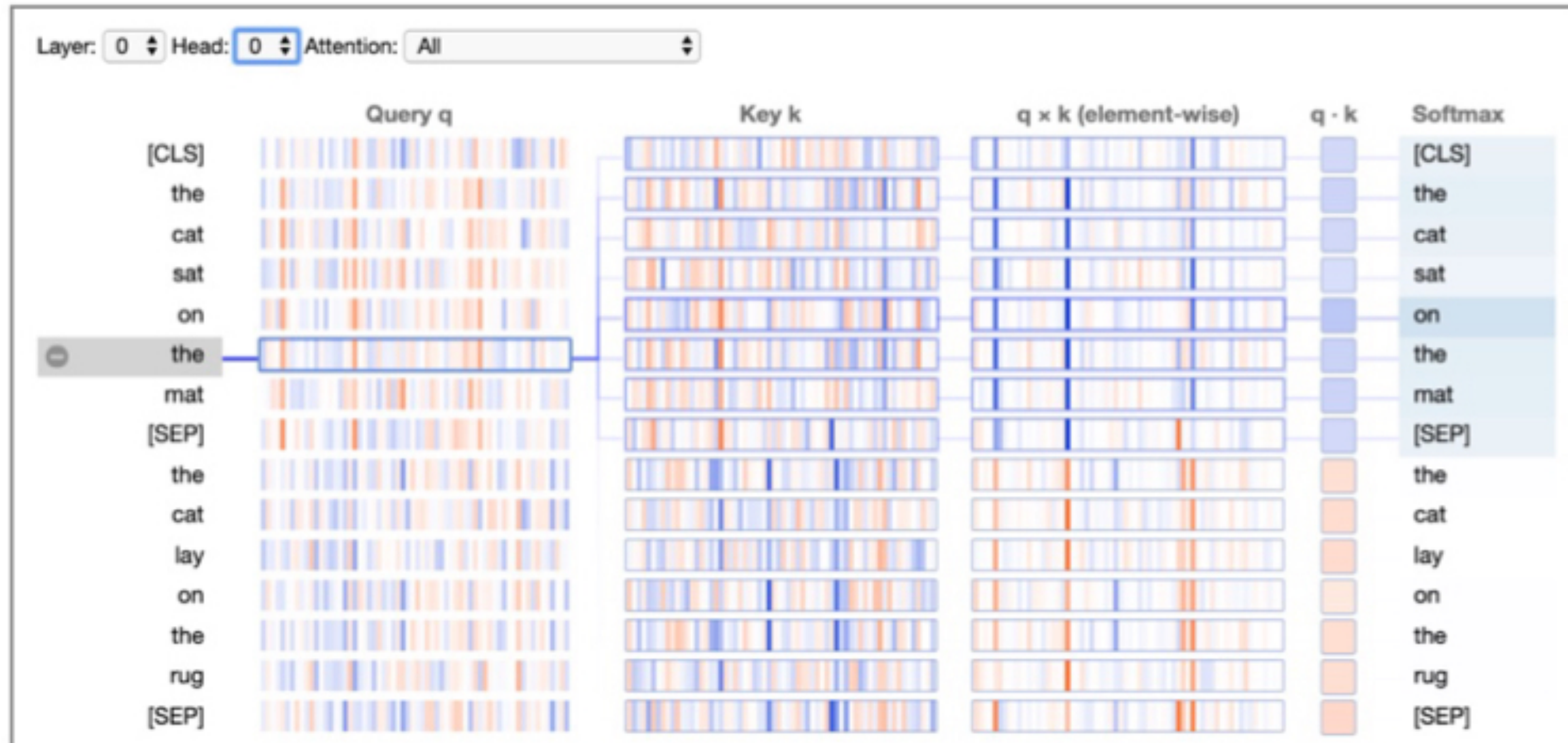


Figure 6: Neuron view of BERT for layer 0, head 0 (same one depicted in Figure 2, left). Positive and negative values are colored blue and orange, respectively, with color saturation based on magnitude of the value. As with the attention-head view, connecting lines are weighted based on attention between the words.

Neuron View

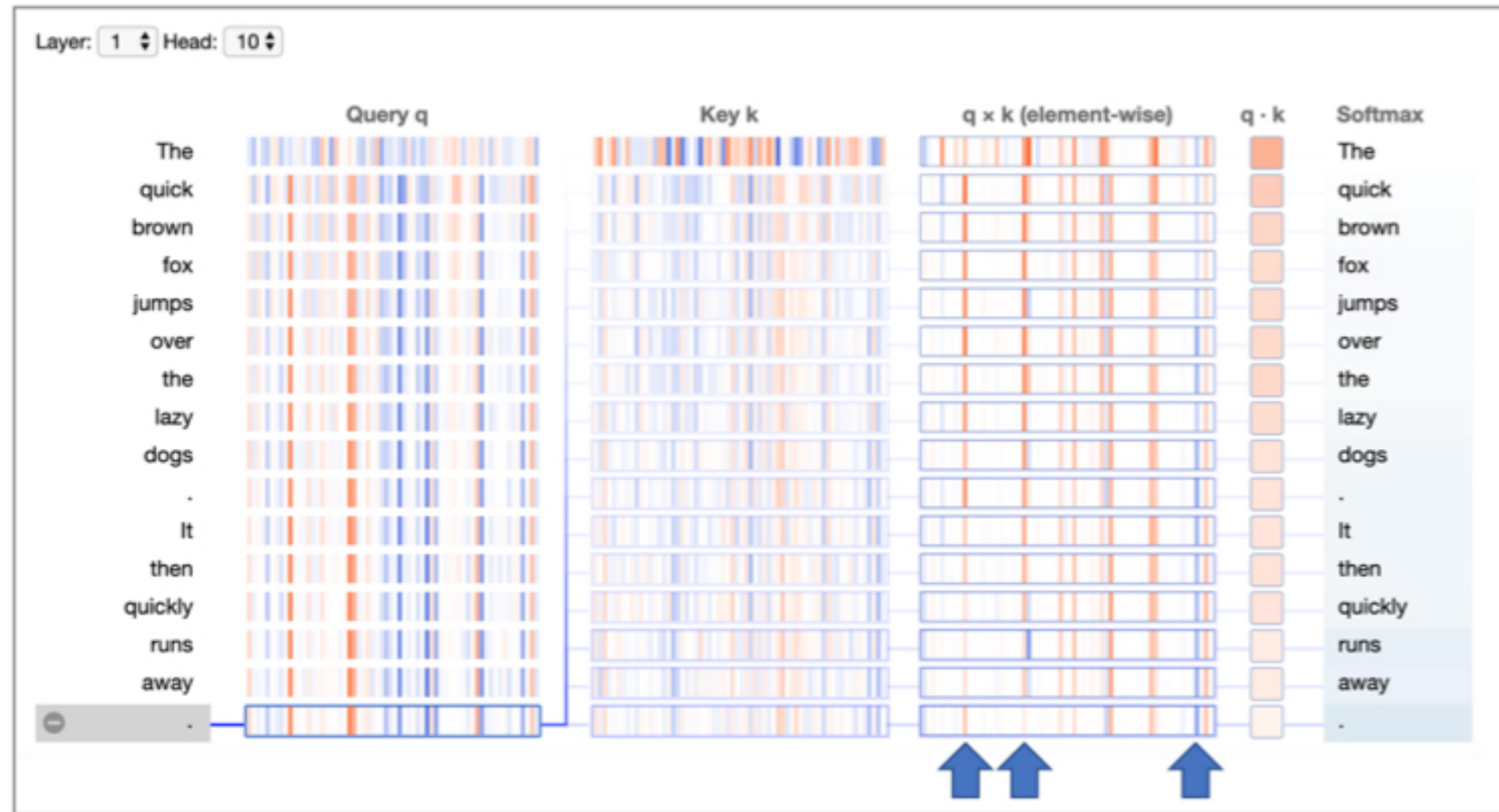


Figure 7: Neuron view of GPT-2 for layer 1, head 10 (same one depicted in Figure 1, center) with last token selected. Blue arrows mark positions in the element-wise products where values decrease with increasing distance from the source token (becoming darker orange or lighter blue).

Neuron View

- The attention weights appear to be largely independent of the content of the input text, based on the fact that all the query vectors have very similar values
- A small number of neuron positions appear to be mostly responsible for this distance-decaying attention pattern

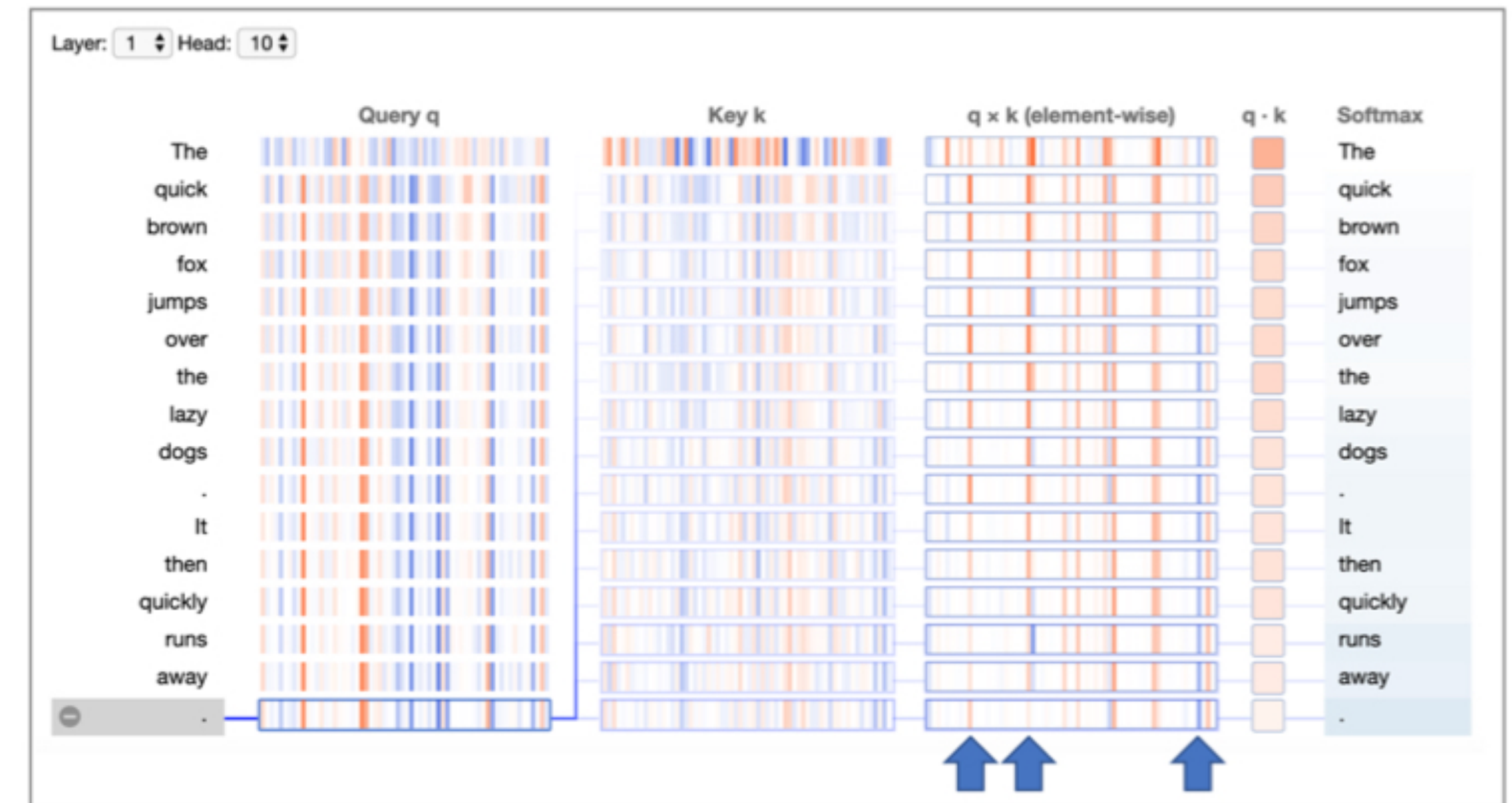


Figure 7: Neuron view of GPT-2 for layer 1, head 10 (same one depicted in Figure 1, center) with last token selected. Blue arrows mark positions in the element-wise products where values decrease with increasing distance from the source token (becoming darker orange or lighter blue).

Use Case

- Model intervention - ex. One might prefer a slower decay rate for a scientific text compared to a children's story. Other heads may afford different types of interventions.

4 Conclusion

- To me, the paper was visually pleasing.
- However, I carefully suggest that it might have been better to give more detailed explanations of how they extracted each weight and computation values.
- I find the tool very useful since it might help understand the blackbox when the model result is somewhat different from what I expect. (plus I already found many many posts explaining Transformer in depth using the site image)

4 Related works

- Llion Jones. 2017. Tensor2tensor transformer visualization
- Interactive visualization and manipulation of attention-based neural machine translation
- Visual interrogation of attention-based models for natural language inference and machine comprehension

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